INFINITE SERIES

Lesson 27

Motivation:

In lesson 26, we have been considering sequences and series with finite (n terms), such as AP, GP and natural numbers with finite terms.

Eg 1:

$$S_n = \sum_{k=1}^{n} [a+d(k-1)] = \underbrace{a}_{T_1} + \underbrace{a+d}_{T_2} + \underbrace{a+2d}_{T_3} + \dots + \underbrace{a+d(n-1)}_{T_n}$$ ← Finite AP series

$$S_n = \frac{n}{2}[2a+d(n-1)]$$ ← Sum of n AP terms

Eg 2:

$$G_n = \sum_{k=1}^{n} ar^{k-1} = \overset{T_1}{a} + \overset{T_2}{ar} + \overset{T_3}{ar^2} + \dots + \overset{T_n}{ar^{n-1}}$$ ← Finite G.P series

$$G_n = \frac{a(1-r^n)}{1-r}$$ ← Sum of n G.P terms.

Eg 3:

$$\sum_{r=1}^{n} r^2 = 1^2 + 2^2 + 3^2 + 4^2 + \dots + n^2$$ ← Finite series of square of natural numbers

$$\sum_{r=1}^{n} r^2 = \frac{n(n+1)(2n+1)}{6}$$ ← Formula sum of n-terms of square of natural no's.

* Finite series such as AP, GP and natural numbers are easy to compute series of n terms. If the number of terms goes to ∞, we obtain infinite series.

Defn: (Infinite Series)

An infinite series is a series whose terms goes to infinity ∞,

denote: (i) $\sum_{n=1}^{\infty} a_n$ or (ii) $\sum_{n=0}^{\infty} b_n$

Eg:

(i) $\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots$ The dots means infinite series

(ii) $\sum_{n=0}^{\infty} 8\left(\frac{1}{2}\right)^n = 8 + 4 + 2 + 1 + \frac{1}{2} + \dots$

CONVERGENCE / DIVERGENCE OF INFINITE SERIES

* An infinite Series $\sum_{n=1}^{\infty} a_n$ can either converge or diverge

Defn: (Convergence)
A series (ie infinite series) is said to be convergent if the sum of n terms tends to a finite value as $n \to \infty$.

ie $\lim_{n\to\infty} S_n = c$, $c \in \mathbb{R}$.
↖ Finite number.

Defn: (Divergence)
If the sum of n terms tends to $\pm\infty$, as $n \to \infty$, we say the infinite series is divergent. ie $\lim_{n\to\infty} S_n = \pm\infty$.

G.P SERIES (Infinite G.P.)

Consider $\sum_{n=1}^{\infty} ar^{n-1} = a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$

Sum of n-terms: $S_n = \dfrac{a(1-r^n)}{1-r}$

$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r} = \frac{a}{1-r} - \lim_{n\to\infty} \frac{ar^n}{1-r} = \begin{cases} \dfrac{a}{1-r} & \text{if } |r|<1 \quad \text{Converges} \\ \pm\infty & \text{if } |r| \geq 1 \quad \text{diverges} \end{cases}$$

* Hence G.P Converges to $\dfrac{a}{1-r}$ if $|r|<1$.

and G.P diverges to $\pm\infty$ if $|r| \geq 1$.

$\therefore S_\infty = \dfrac{a}{1-r}$ if $|r|<1$ (sum to infinity of a G.P) Converges

and $S_\infty = \pm\infty$ if $|r| \geq 1$. (diverges)

A.P Series

* All AP series are divergent as $n \to \infty$.

Since $\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{n}{2}\left[2a + d(n-1)\right] = \infty$ (diverges).

All infinite AP series are divergent.

Example 1

State whether the following series are Convergent / divergent.

(a) $\sum_{n=1}^{\infty}(5-3n)$ (b) $\sum_{n=1}^{\infty}8\left(-\frac{3}{4}\right)^n$ (c) $\sum_{n=1}^{\infty}\left(\frac{5}{3}\right)^n$ (d) $\sum_{n=1}^{50}(2n+1)$

Solution

(a) Diverges to $-\infty$. Since is an infinite AP series.

(b) Converges to $S_\infty = \frac{a}{1-r} = \frac{8(-3/4)}{1-(-3/4)} = \frac{-6}{7/4} = -\frac{24}{7}$, since G.P $|r| = \frac{3}{4} < 1$.

(c) Diverges to ∞, Since it is a G.P with $|r| = \frac{5}{3} > 1$.

(d) Converges, Since it is a finite AP.

Note: * All finite series are Convergent, since they terminate.

* Eg: (i) $S = 1^3 + 4^3 + 7^3 + 10^3 + \dots + 178^3$ ← Terminates hence

(ii) $T = 2 + 5 + 8 + 11 + \dots + 1007$

** All infinite G.P series are Convergent provided $|r| < 1$

Eg: (i) $G = \sum_{n=1}^{\infty} 10\left(-\frac{3}{5}\right)$ ← Here $|r| = \frac{3}{5} < 1$

(ii) $H = \sum_{n=0}^{\infty} 5\left(\frac{2}{3}\right)^n$ ← $|r| = \frac{2}{3} < 1$

Convergent

** All infinite G.P series are divergent provided $|r| \geq 1$

Eg (i) $\sum_{n=1}^{\infty} 5\left(-\frac{3}{2}\right)^{n-1}$ ← Here $|r| = \frac{3}{2} > 1$

(ii) $\sum_{n=0}^{\infty} 10\left(-\frac{5}{3}\right)^n$ ← $|r| = \frac{5}{3} > 1$

divergent